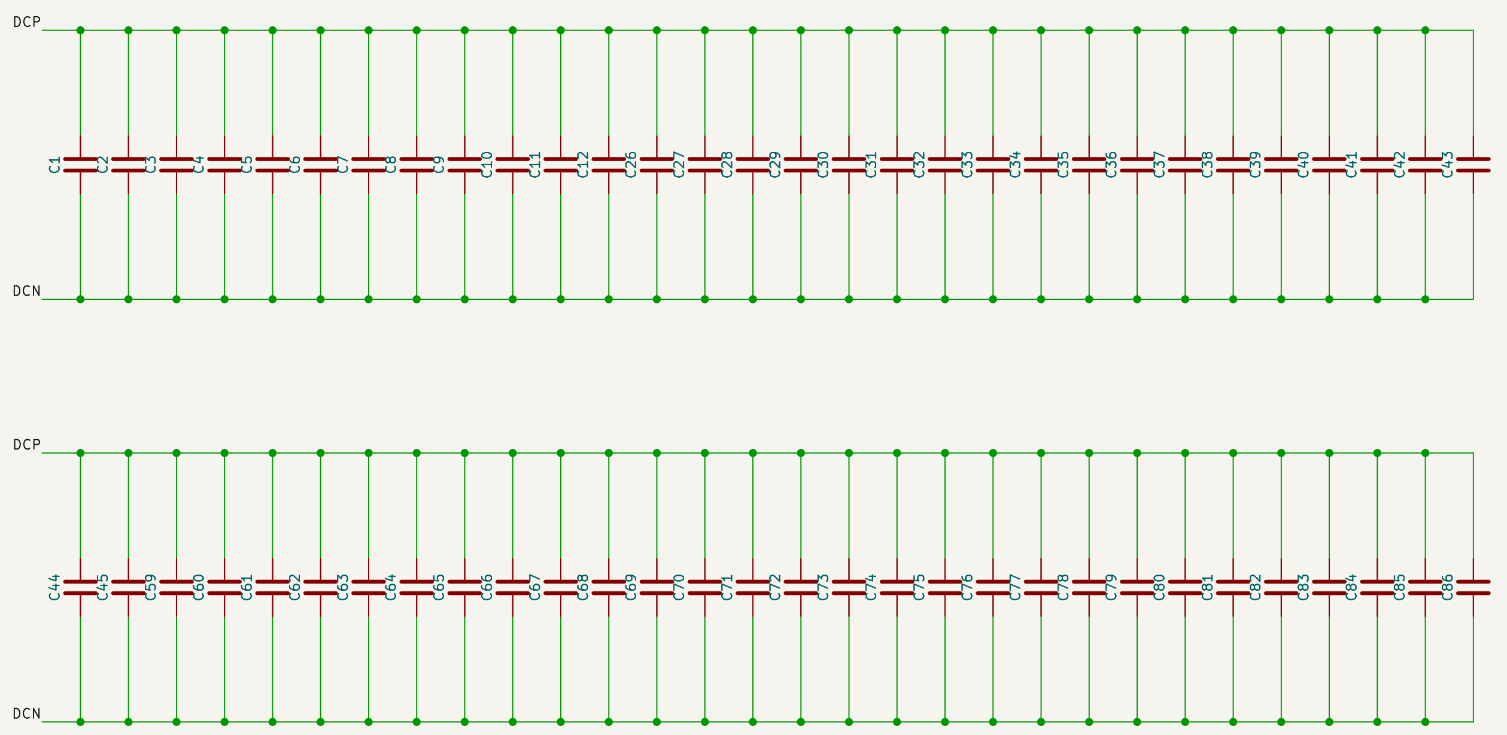
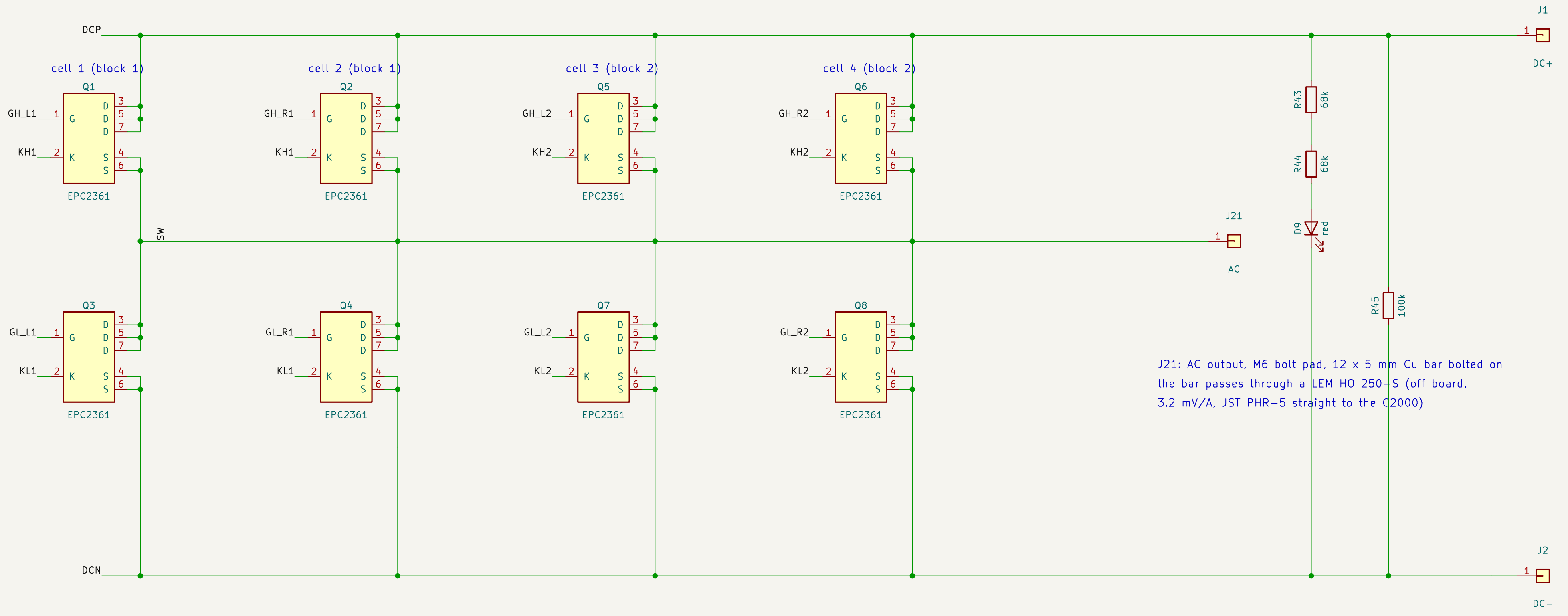
POWER STAGE AND DC LINK (block 1: Q1–Q4, block 2: Q5–Q8; one cell = HS + LS transistor, LS source straight to DC−; AC of all four cells joined on the PCB)



DC link, 60 capacitors: 40 x 1uF 100V X7S, 20 x 22uF 100V X7S 2220 (1 uF 0805: bypass at the HS drains of each cell, top + bottom; 22 uF 2220: bank)



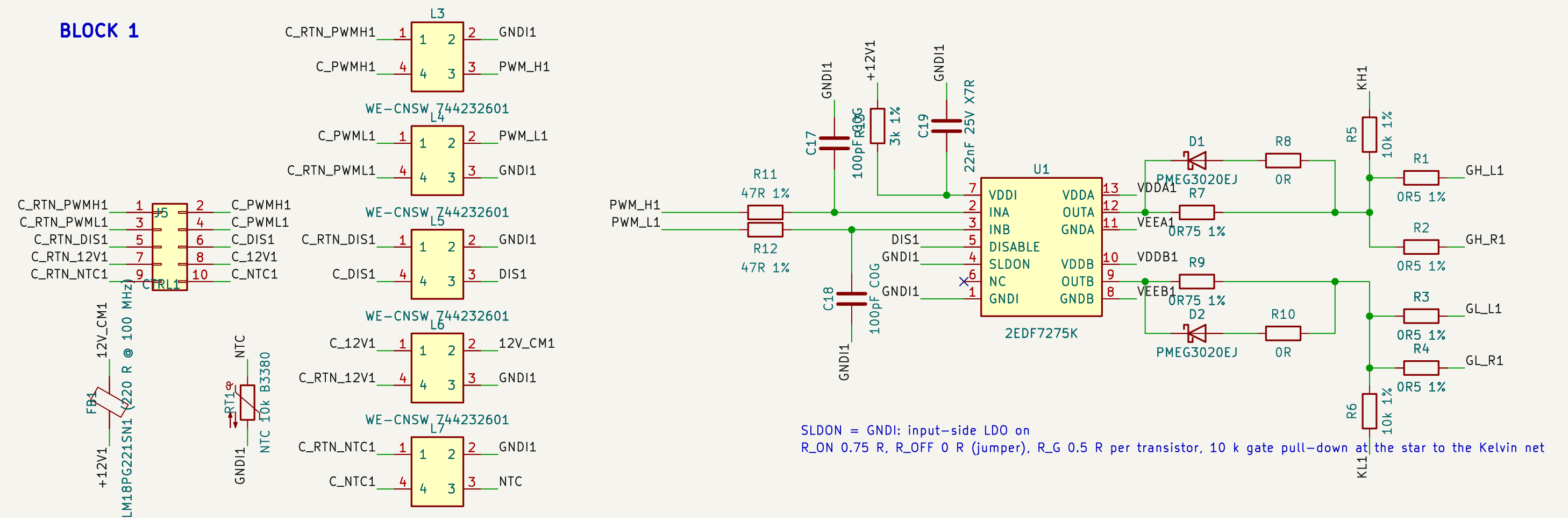
J21: AC output, M6 bolt pad, 12 x 5 mm Cu bar bolted on
the bar passes through a LEM HO 250–S (off board,
3.2 mV/A, JST PHR–5 straight to the C2000)

No shunt (load–test board): LS source stripes via–in–pad (blind L1–L2) into the DC− plane

DC link present LED, R45 100 k bleeder

GATE DRIVER BLOCKS: control header with common–mode chokes, 2EDF7275K input filters, R_ON / D + R_OFF, star, R_G per transistor (block 1 drives Q1–Q4, block 2 drives Q5–Q8)

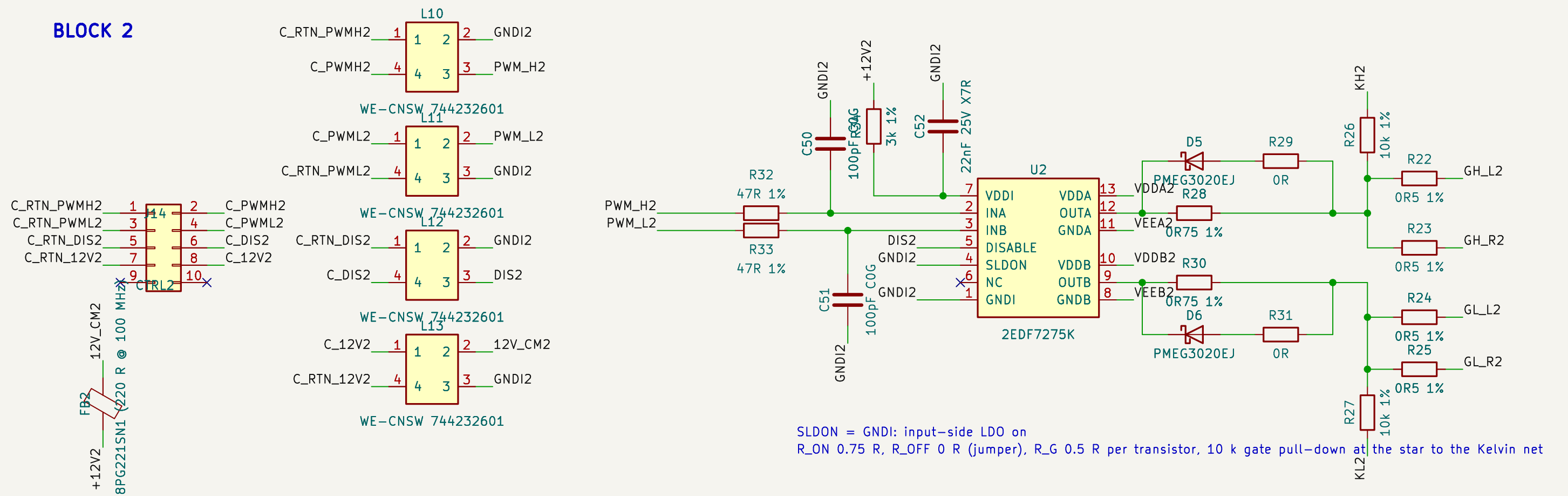
BLOCK 1



SLDON = GNDI: input–side LDO on
R_ON 0.75 R, R_OFF 0 R (jumper), R_G 0.5 R per transistor, 10 k gate pull–down at the star to the Kelvin net

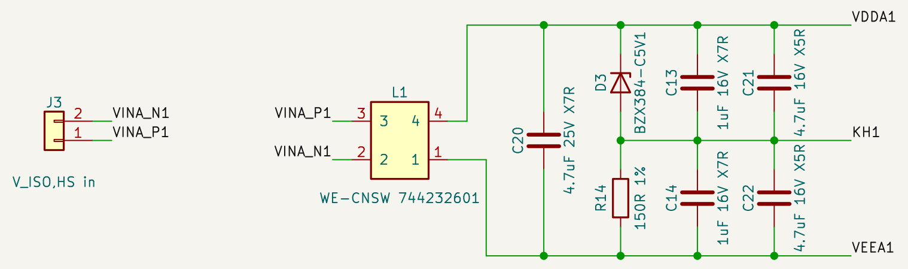
NTC on the control side (board temperature at the driver)

BLOCK 2

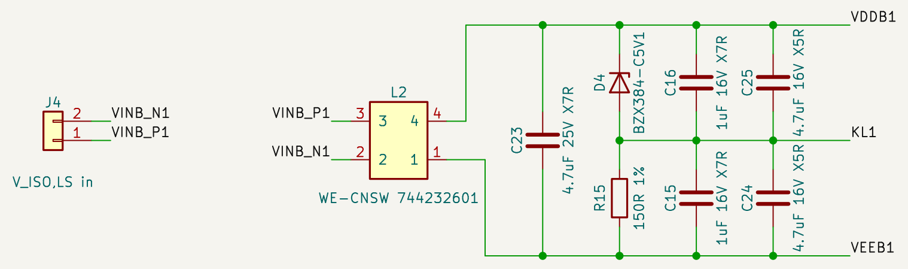


SLDON = GNDI: input–side LDO on
R_ON 0.75 R, R_OFF 0 R (jumper), R_G 0.5 R per transistor, 10 k gate pull–down at the star to the Kelvin net

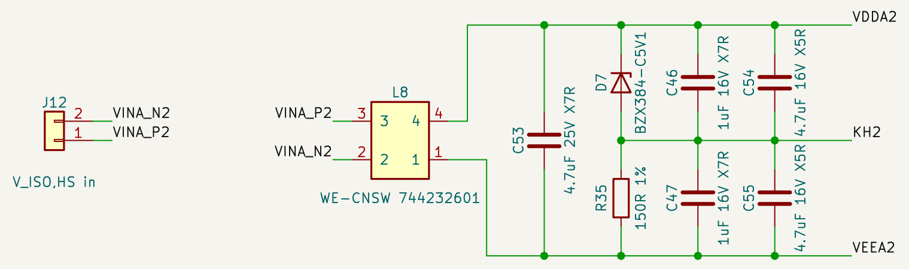
GATE SUPPLY ISLANDS: external isolated supply (one HS, one LS, split per block) –> WE–CNSW –> 5.1 V Zener VDD–S, 150 R bias S–VEE: V_GS,on = +5.1 V, V_GS,off = −(V_ISO − 5.1 V)



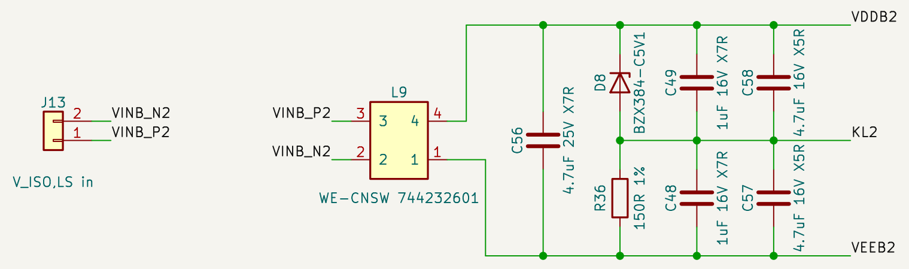
J3 / L1: HS block 1



J4 / L2: LS block 1

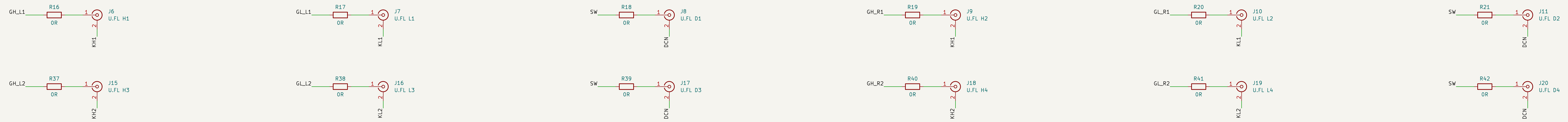


J12 / L8: HS block 2



J13 / L9: LS block 2

U.FL MEASUREMENT PORTS (bottom side, at each cell): V_GS HS / LS (0 R, shield = Kelvin), V_DS LS (4.99 k 2512 series into the 50 R scope input = 100:1, 1.1 W at 75 V; shield = LS source = DC−)



MOUNTING: H1–H4 board corners (insulated), H5–H7 heatsink spreader; heatsink HS1 (Fischer LAM 4, 100 mm, rotated 90 deg, tube across the board width) is a board–only item

Nets: DCP = DC+, DCN = DC−, SW = joined switch node of all four cells. KH1/KH2 and KL1/KL2 are the Kelvin source nets (pad 2 of each EPC2361); they meet the power source only inside the package.

LS power source = DC− directly (no shunt). The LS gate drive and the LS supply are referenced to the Kelvin net KLx, not to DC−.

Block suffix 1/2 on every driver–side net (GNDI1, VDDA2, ...). Control side (CTRLx 2 x 5 header): PWM_H, PWM_L, DIS, 12 V and (block 1) NTC each with their own return C_RTN_* through one WE–CNSW common–mode choke per pair; 12 V then through a ferrite bead. The control side is a top–layer–only area with a 1200 V barrier (4 mm outer, 3 mm L2) to all power copper; see reports/pcb–rules/PCB_rules_and_checklist.md.

No short–circuit protection on this board. Closed–loop control from a TI TMS320F28379D over the CTRL headers.

generated by scripts/make_kicad_sch_revb.py; netlist identical to the PCB (check_sch_vs_pcb.py)
4 x EPC2361 per switch, 2 x 2EDF7275K, no shunt, LAM 4 heatsink, external isolated supplies, U.FL p
KTH
Sheet: /
File: spb–bb–revb–load.kicad_sch
Title: SPB leg, Rev B, load–test board
Size: A1
Date: 2026–10–14
Rev: B
KiCad E.D.A. kicad–cli 7.0.10
Id: 1/1